Session Objectives

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1. Describe, discuss, and explain the pathogenesis, morphological features (gross and microscopic), and major clinical correlations of **acute inflammatory** dermatoses which include **urticaria, acute eczematous dermatitis, erythema multiforme, Stevens Johnson syndrome**
2. Describe, discuss, and explain the pathogenesis, morphological features (gross and microscopic), and major clinical correlations of **chronic inflammatory** dermatoses which include **psoriasis** and **lichen planus**
3. Describe, discuss, and explain the pathogenesis, morphological features (gross and microscopic), and major clinical correlations of major **bullous** diseases which include **pemphigus vulgaris, bullous pemphigoid,** and **dermatitis herpetiformis**
4. Describe, discuss, and explain the pathogenesis, morphological features (gross and microscopic) and major clinical correlations of **acne vulgaris and rosacea**
5. Describe, discuss, and explain the pathogenesis, morphological features (gross and microscopic) and major clinical correlations of **erythema nodosum (panniculitis)**

More skin pathology terminology

Microscopic Feature	Description
Acanthosis	Diffuse epidermal hyperplasia
Dyskeratosis	Abnormal, premature keratinization within cells below the stratum granulosum
Hyperkeratosis	Thickening of the stratum corneum, often associated with a qualitative abnormality of the keratin
Lentiginous	Linear patter of melanocyte proliferation within the epidermal basal cell layer
Parakeratosis	Keratinization with retained nuclei in the stratum corneum
Spongiosis	Intracellular edema of the epidermis

Robbins and Cotran, Pathologic Basis of Disease, 10th ed., 2020, Ch. 25, Table 25.1

Review of hypersensitivity reactions

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- **Type I** (immediate) – **IgE** binds and activates mast cells (ex. anaphylaxis, asthma, urticaria)

- **Type II** (cytotoxic) – **IgG/IgM** antibody dependent, bind to antigen (complement, opsonization/phagocytosis, natural killer (NK) cells bind Ab Fc region, cell dysfunction – hyperthyroidism in Graves' disease, etc.), other examples: Incompatible RBC transfusion reaction, Goodpasture syndrome

- **Type III** (immune complex) – **Ag-Ab complexes** deposition in tissues causing inflammation (ex. lupus, rheumatoid arthritis)

- **Type IV** (delayed type) – cell mediated toxicity, **T cell** mediated, 24-72 hours to develop, antibody independent reactions (ex. contact dermatitis, tuberculin skin test)

Acute inflammatory dermatoses

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1. Urticaria
2. Acute eczematous dermatitis
3. Erythema multiforme

Urticaria (hives): Overview

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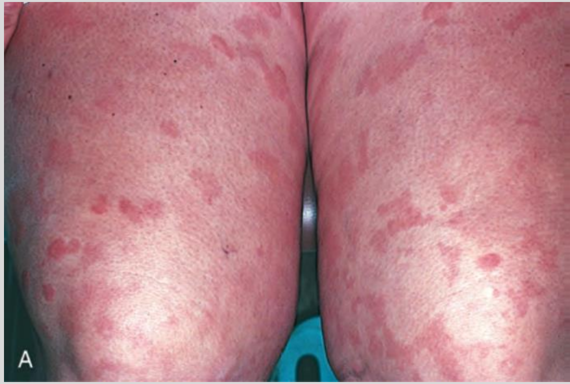
- **Localized mast cell degranulation**
- **Histamine** release, activates H1 & H2 receptors → ↑ vascular permeability, vasodilation → **wheals** + **pruritis**
- Affect **all ages**
- Commonly on trunk, extremities, face
- Majority of patients reveal **no underlying cause**
- Lesions develop & fade → hours/a day
- Can recur → **cycles** can last days/months



Source: Robbins and Cotran, Pathologic Basis of Disease, 10th edition, 2020, Fig. 25.21

Urticaria: Pathogenesis

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- **Mast cell-dependent, IgE-dependent:** Post-exposure to antigens → **Type I** Hypersensitivity reaction
  - When systemic → **anaphylaxis**
- **Mast cell-dependent, IgE-independent:** Substances **directly trigger mast cell** degranulation (examples: Opiates, certain antibiotics, contrast media)
- **Mast cell-independent, IgE-independent: Local factors** that ↑ vascular permeability w/o degranulating mast cells
  - **Aspirin** mechanism is unknown
  - **Hereditary Angioneurotic edema:** Rare, potentially life threatening genetic mutation w/ **deficiency of C1 esterase inhibitor** → ↑bradykinin (vasoactive mediator)

Urticaria: Histopathology

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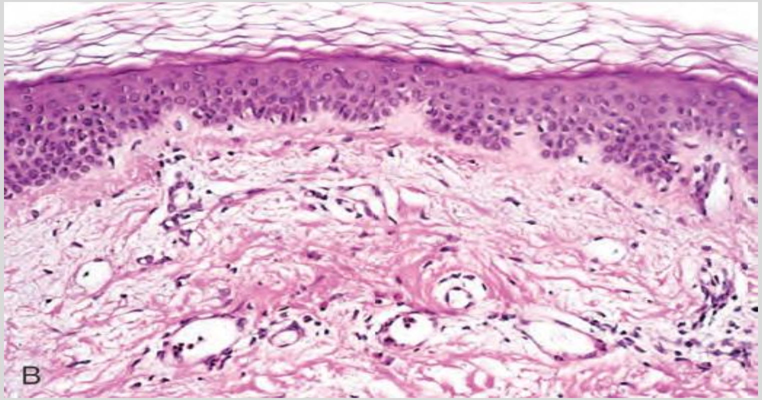

Robbins and Cotran, Pathologic Basis of Disease, 10th edition, 2020, Fig. 25.21

- **Dermal edema** - spacing of dermal collagen bundles
- Sparse superficial perivenular mononuclear cells, possibly eosinophils & neutrophils
- **Dermal lymphatics dilated** - edema fluid
- **No changes in epidermis**

Allergic contact dermatitis

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- **Type IV** hypersensitivity
- Triggered by **contact allergens** (ex. nickel, poison ivy, fragrances, latex)
- Affects **superficial dermis**, mononuclear inflammatory reaction w/o eosinophils

- Dermatology: An Illustrated Colour Text, Sixth Edition
- **Watchband** (clasp's metal alloy) etiology

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Additional causes of eczema

- **Irritant contact dermatitis**
  - Direct chemical irritation (detergents, acids) produce **non-immunologic**, dose-dependent reaction
- **Photo-eczematous dermatitis**
  - UV light, double-edged sword for eczema
    - **Trigger for flares**, excessive exposure can damage the skin barrier
    - **Rx** in controlled doses - (UVB) reduces inflammation

Robbins and Cotran, Pathologic Basis of Disease, 10th edition, 2020, Fig. 25.23

Dermatology: An Illustrated Colour Text, Sixth Edition

Erythema multiforme: Overview

- Rare, **self-limiting** hypersensitivity
  - Reaction to certain **infections & drugs**
- Affects **all ages**
- Mediated by **T-cells** (both CD4+ and CD8+ T cells) → **Type IV** delayed reaction
- Triggers
  - **HSV** (most common), Mycoplasma pneumoniae, EBV, CMV, VZV, histoplasmosis
  - Sulfa drugs (**TMP-SMX**), NSAIDS, penicillins, barbiturates, phenytoin
  - Malignancies, autoimmune disease (SLE, IBD)

Stevens-Johnson Syndrome (SJS) & TEN

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- **Severe mucocutaneous** reaction
  - **Widespread epidermal necrosis & detachment**
- Often triggered by **meds or infections** (ex. **sulfa** drugs most commonly)
- Clinically: Lesions involve skin and lips/oral **mucosa**, conjunctiva, urethra, genital areas
  - Initially erythematous macules → bullae → epidermal detachment **(+ Nikolsky sign**)
  - **>10% skin** involvement
  - If >**30%, renamed Toxic Epidermal Necrolysis (TEN)**
- Management→ Stop offending drug + supportive care
- Prognosis **-> 5-10% mortality** for SJS, **>30% for TEN**

https://my.clevelandclinic.org/health/diseases/17656-stevens-johnson-syndrome

Psoriasis: Clinical

- Lesion appearance varies: Typically well-demarcated, pink to **salmon plaques** covered by loosely adherent **silver-white scale**
- **Nail changes**
- **Koebner Phenomenon**
- **Auspitz sign**

Robbins and Cotran, Pathologic Basis of Disease, 10th edition, 2020, Fig. 25.25

Lichen Planus: Overview

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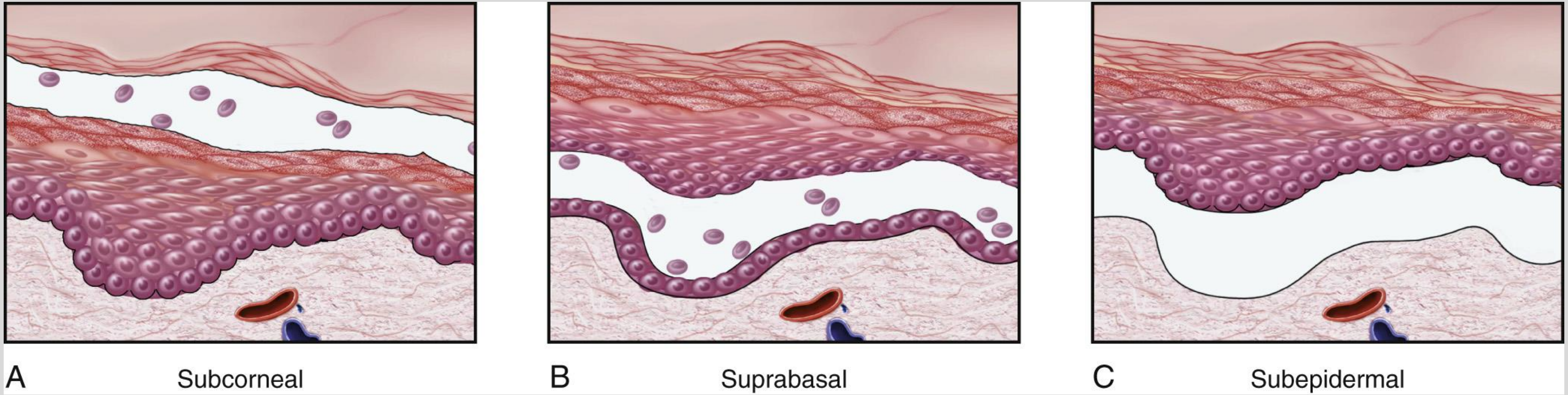
- **Pruritic, purple, polygonal, planar, papules, plaques** that affects skin, mucosa, nails, hair follicles

- Typically **middle aged**
- **Self-limiting** → resolves spontaneously over course of a few years
- Pathogenesis not fully known
  - Possibly CD8+ T-cell response to antigens in basal epidermal or dermo-epidermal junction
- Associated w/ **Hepatitis C**, although **most** cases **idiopathic**

Bullous (blistering) disorders

1. Pemphigus Vulgaris
2. Bullous pemphigoid
3. Dermatitis herpetiformis
4. Epidermolysis bullosa

Type of blister helps determine the cause



Stratum corneum forms the roof of the bulla (ex. friction blister, pemphigus foliaceus)

A portion of epidermis, including the stratum corneum, forms the roof (ex. **pemphigus vulgaris**)

Entire epidermis separates from the dermis (ex. **bullous pemphigoid**)

Robbins and Cotran, Pathologic Basis of Disease, 10th ed., 2020, Ch. 25, Fig. 25.27

Pemphigus vulgaris: Clinical findings

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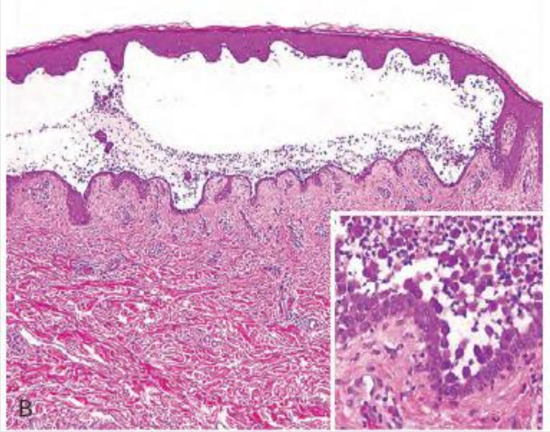
- Inflammatory, blistering skin diseases caused by **autoantibodies (IgG)**
- Auto-abs **disrupt intercellular keratinocyte attachments** within epidermis & mucosal epithelium
- Patients in **30s-50s** & males, females equally effected
- Usually **benign**, but extreme variants **can be fatal** w/o Rx
- Involved **skin & mucosa** (**oral** ulcers may present months **before skin** lesions)
- Lesions are superficial vesicles and blisters rupture easily
    - (+ **Nikolsky sign)**
- Treat with immunosuppressive agents → ↓ Abs

Robbins and Cotran, Pathologic Basis of Disease, 10th edition, 2020, Ch. 25
https://www.nature.com/articles/s41415-024-7074-8
https://ucfhealth.com/our-services/dermatology/pemphigus-vulgaris/

Pemphigus vulgaris: Histopathology

- Autoimmune: **IgG** autoantibodies against **desmogleins** (subunit of desmosome)
- Causes **suprabasal** blisters
- **Acantholysis**: Dissolution of intercellular bridges immediately above basal cell layer
- **Row of tombstones**: Blister leaves behind single layer of intact basal cells



Robbins and Cotran, Pathologic Basis of Disease, 10th edition, 2020, Fig. 25.29

Pemphigus vulgaris: Histopathology - direct immunofluorescent microscopy detects epidermal autoantibodies

- Distinct **net-like** pattern of intercellular IgG deposits
- IgG is usually seen at all levels of the epithelium
- Distribution of **desmoglein 1 & 3**

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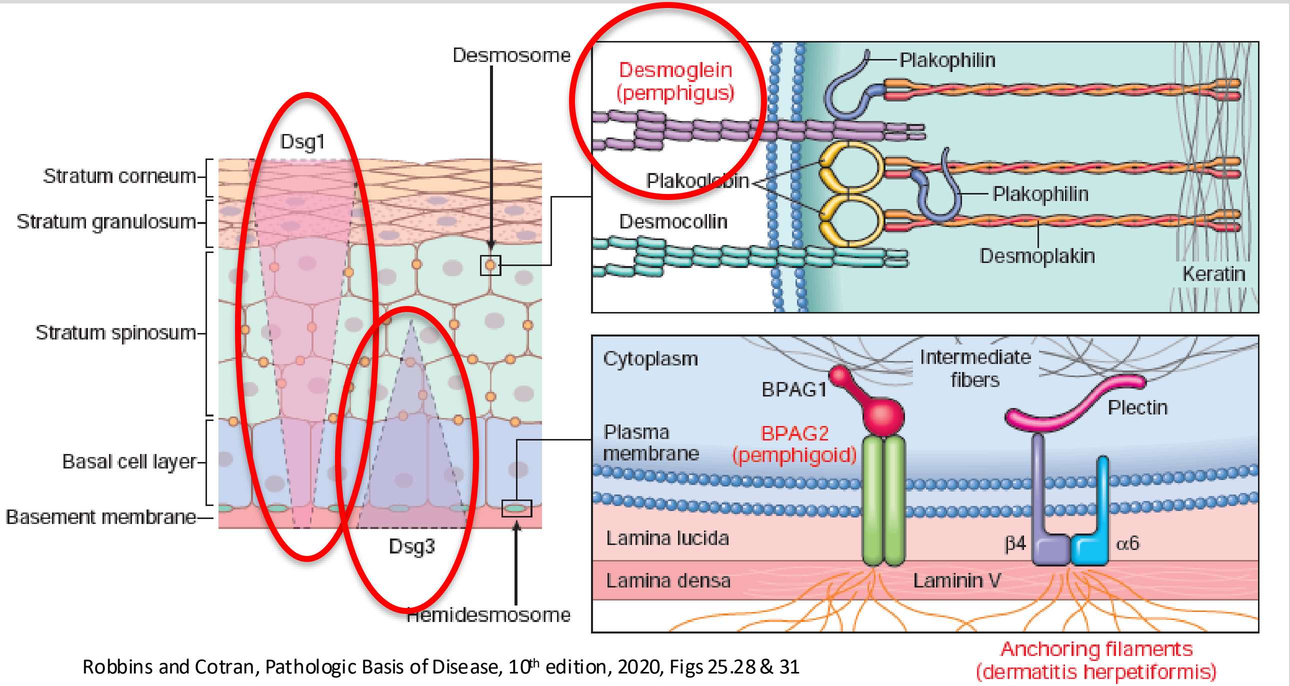

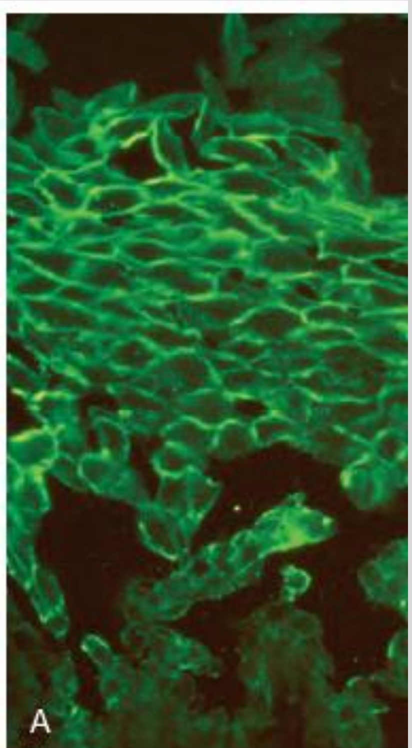

Robbins and Cotran, Pathologic Basis of Disease, 10th edition, 2020, Figs 25.28 & 31

Bullous pemphigoid: Pathogenesis

- **IgG autoantibodies** against **BPAG2** - part of **hemidesmosomes** (attach basal keratinocytes to the basement membrane)
- Forms **subepidermal blisters** (lamina lucida), nonacantholytic

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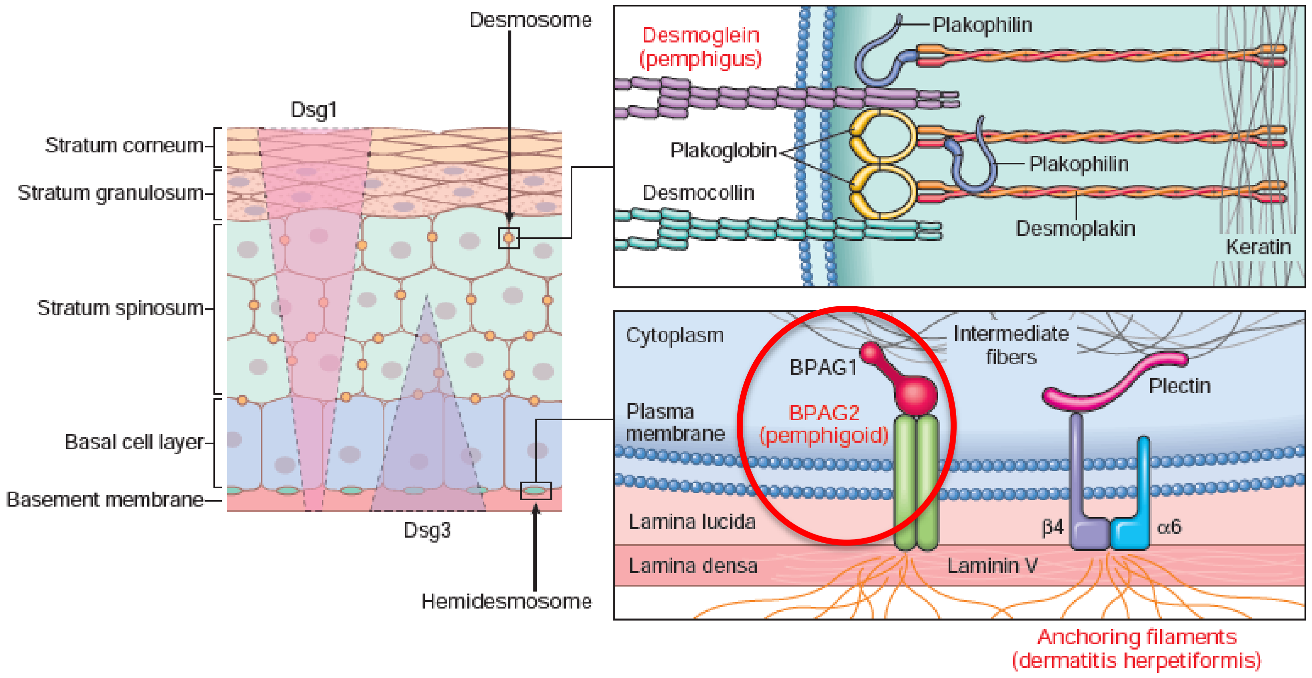
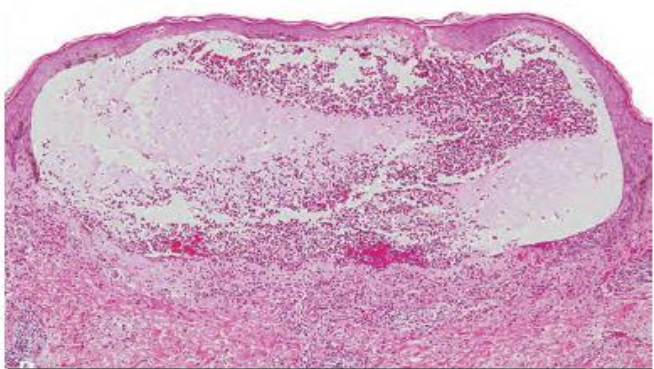
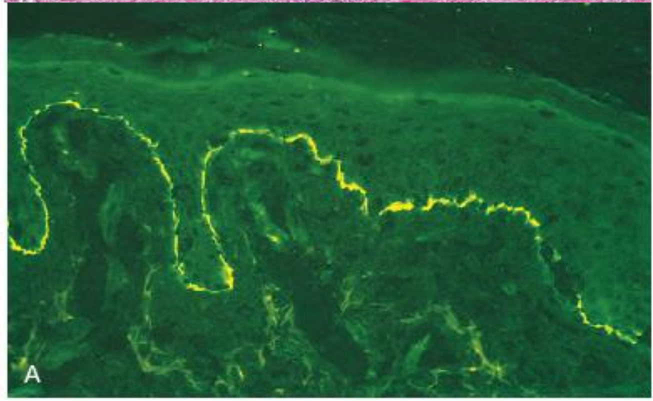


Robbins and Cotran, Pathologic Basis of Disease, 10th edition, 2020, Figs 25.28

Bullous pemphigoid: Histopathology

- **Sub-epidermal & non-acantholytic**
- **Superficial dermal edema**
- Basal layer vacuolization, leading to fluid-filled blisters
- Full thickness of epidermis serving as roof of blister
    - **- Nikolsky sign**
- Direct immunofluorescent microscopy detects **subepidermal autoantibodies** - linear, continuous IgG deposition along the dermo-epidermal junction (hemidesmosomes are here)

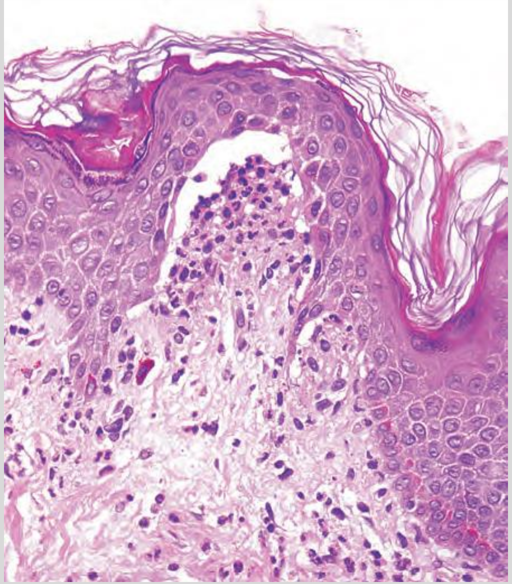


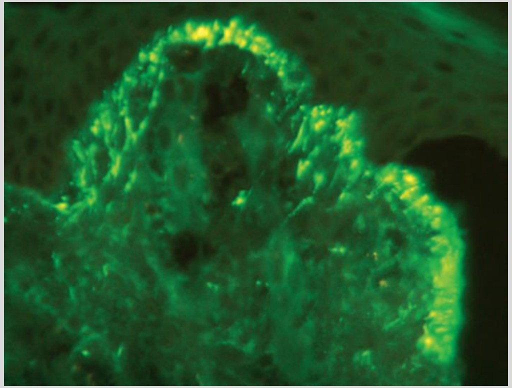


Robbins and Cotran, Pathologic Basis of Disease, 10th edition, 2020, Fig. 32 & 33

Dermatitis herpetiformis: Histopathology

- **Microabscesses**: Fibrin and **neutrophils** accumulate at **tips of dermal papillae**
- Basal cell layer vacuolization and dermo-epidermal junction separation → **subepidermal blisters**
- Direct immunofluorescent microscopy detects a **discontinuous**, **granular IgA** deposits at tips of dermal papillae (anti-tissue transglutaminase (tTG) and anti-epidermal transglutaminase (eTG))





Robbins and Cotran, Pathologic Basis of Disease, 10th edition, 2020, Fig. 25.34

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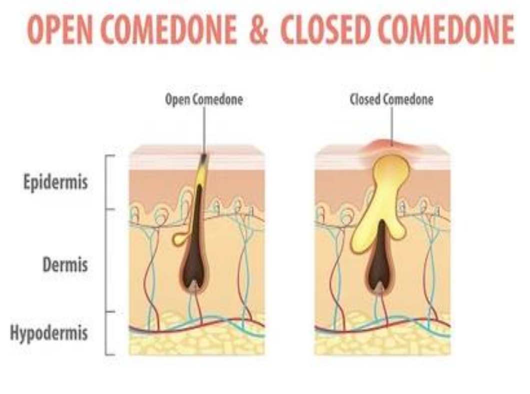
Acne vulgaris: Clinical findings

- Virtually universal in the middle to late teenage years
- Incidence: M=F, males typically more severe
- Exacerbated by
  - Drugs – **corticosteroids**, **testosterone**, gonadotropins, **contraceptives**
  - **Occlusion** of sebaceous glands – heavy clothing, cosmetics, chin straps in football
- Can be familial - possibly hereditary
- **Noninflammatory** acne - both types can coexist
  - **Open comedones** (blackheads): Central black **keratin plug** - oxidation of melanin
  - **Closed comedones** (whiteheads): Keratin plug is trapped beneath the epidermal surface
- **Inflammatory** acne - both types can coexist
  - **Erythematous** papules, nodules, pustules
  - **Severe variants** can result in sinus tract formation and **scarring**
- Histopathology shows variable infiltrates of lymphocytes and macrophages present in and around follicles

https://doctorlib.info/medical/fitzpatrick-atlas-dermatology/4.html



Acne vulgaris: Pathogenesis, treatments

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- Likely multifactorial causes
  1. Keratinization of lower portion of follicular stalk and development of **keratin plug** - that blocks outflow of sebum to the skin surface
  2. Hypertrophy of sebaceous glands - under the influence of **androgens** during **puberty**
  3. Lipase-synthesizing bacteria (**Propionibacterium acnes**) colonize hair follicle converting lipids within sebum to proinflammatory fatty acids
  4. Secondary inflammation of involved follicle
- Rxs: Antibiotics for elimination of bacteria & isotretinoin has strong anti-sebaceous action